

Lecture 2: Functions of Several Variables

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- 1 Partial Derivatives
- 2 The Total Derivative
- 3 Explicit Functions from $\mathbb{R}^n$ to $\mathbb{R}^m$
- 4 The Chain Rule
- 5 Directional Derivative
- 6 Higher Order Derivatives
- 7 Implicit Function Theorem

- ▶ Consider a function $y = f(x_1, x_2, \dots, x_n)$, where each x_i can vary without affecting the others.
- ▶ If x_i undergoes a change Δx_i while all other x_j s remain fixed, there will be a change in y , Δy .
- ▶ The partial derivative of f with respect to x_i :

$$\frac{\partial f}{\partial x_i}(x_1^0, \dots, x_i^0, \dots, x_n^0) = \lim_{h \rightarrow 0} \frac{f(x_1^0, \dots, x_i^0 + h, \dots, x_n^0) - f(x_1^0, \dots, x_i^0, \dots, x_n^0)}{h}$$

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Based on the definition of partial derivatives:

$$\blacktriangleright F(x^* + \Delta x, y^*) - F(x^*, y^*) \approx \frac{\partial F}{\partial x}(x^*, y^*) \Delta x$$

$$\blacktriangleright F(x^*, y^* + \Delta y) - F(x^*, y^*) \approx \frac{\partial F}{\partial y}(x^*, y^*) \Delta y$$

$$\blacktriangleright F(x^* + \Delta x, y^* + \Delta y) - F(x^*, y^*) \approx \frac{\partial F}{\partial x}(x^*, y^*) \Delta x + \frac{\partial F}{\partial y}(x^*, y^*) \Delta y$$

Linear approximation:

- ▶ $F(x^* + \Delta x, y^* + \Delta y) \approx F(x^*, y^*) + \frac{\partial F}{\partial x}(x^*, y^*)\Delta x + \frac{\partial F}{\partial y}(x^*, y^*)\Delta y$
- ▶ The “tangent plane” can be written as the parametric equation

$$(x^*, y^*, F(x^*, y^*)) + s \left(1, 0, \frac{\partial F}{\partial x}(x^*, y^*) \right) + t \left(0, 1, \frac{\partial F}{\partial y}(x^*, y^*) \right)$$

- ▶ We use *differentials*: dF, dx, dy , to denote the variations on the tangent plane
- ▶ *Total differential*: $dF = \frac{\partial F}{\partial x}(x^*, y^*)dx + \frac{\partial F}{\partial y}(x^*, y^*)dy$
- ▶ Example: $h = x^3 \ln y$,

$$dh = 3x^2 \ln y dx + \frac{x^3}{y} dy$$

Generally, for $y = F(x_1, x_2, \dots, x_n)$

- ▶ *Total differential*: $dF = \frac{\partial F}{\partial x_1}(\mathbf{x}^*)dx_1 + \frac{\partial F}{\partial x_2}(\mathbf{x}^*)dx_2 + \dots + \frac{\partial F}{\partial x_n}(\mathbf{x}^*)dx_n$
- ▶ Jacobian derivative of F at $\mathbf{x}^*$:

$$DF_{\mathbf{x}^*} = \left(\frac{\partial F}{\partial x_1}(\mathbf{x}^*), \frac{\partial F}{\partial x_2}(\mathbf{x}^*), \dots, \frac{\partial F}{\partial x_n}(\mathbf{x}^*) \right)$$

- ▶ It means $dF = DF_{\mathbf{x}^*} \cdot d\mathbf{x}$ where

$$d\mathbf{x} = (dx_1, dx_2, \dots, dx_n)'$$

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- Functions with several endogenous variables: $F = (f_1, f_2, \dots, f_m) : \mathbb{R}^n \rightarrow \mathbb{R}^m$.
- For each $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$

$$f_1(x^* + \Delta x) - f_1(x^*) \approx \frac{\partial f_1}{\partial x_1}(x^*)\Delta x_1 + \dots + \frac{\partial f_1}{\partial x_n}(x^*)\Delta x_n$$

$$f_2(x^* + \Delta x) - f_2(x^*) \approx \frac{\partial f_2}{\partial x_1}(x^*)\Delta x_1 + \dots + \frac{\partial f_2}{\partial x_n}(x^*)\Delta x_n$$

$$\vdots$$

$$f_m(x^* + \Delta x) - f_m(x^*) \approx \frac{\partial f_m}{\partial x_1}(x^*)\Delta x_1 + \dots + \frac{\partial f_m}{\partial x_n}(x^*)\Delta x_n$$

- Use vector and matrix notation:

$$F(\mathbf{x}^* + \Delta \mathbf{x}) - F(\mathbf{x}^*) \approx \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}^*) \end{pmatrix} \begin{pmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{pmatrix}$$

Jacobian Derivative

We call the matrix

$$DF(\mathbf{x}^*) = F'(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial f_m}{\partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

the **Jacobian derivative** of F at $\mathbf{x}^*$.

Example: In a two-commodity world, consider the pair of constant-elasticity demand functions: $Q_1 = 6p_1^{-2}p_2^{3/2}y$ and $Q_2 = 4p_1p_2^{-1}y^2$ when $p_1^* = 6$, $p_2^* = 9$ and $y^* = 2$.

$$dQ_1 = -3dp_1 + 1.5dp_2 + 4.5dy$$

$$dQ_2 = \frac{16}{9}dp_1 - \frac{32}{27}dp_2 + \frac{32}{3}dy$$

If both prices rise by 0.1 and income falls by 0.1, then $dQ_1 = -0.6$ and $dQ_2 \approx -1$. You can verify this calculation in matrix notation.

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- ▶ We define a curve in $\mathbb{R}^n$ by $\mathbf{x}(t) = (x_1(t), x_2(t), \dots, x_n(t))$, where each x_i is a continuous map from $\mathbb{R}$ to $\mathbb{R}$.
- ▶ $x_i(t)$ are **coordinate functions** and t is the parameter describing the curve.
- ▶ Examples:

- ▶ The line segment connecting $(0, 0)$ and $(1, 1)$ can be parameterized as

$$x(t) = t, y(t) = t, 0 \leq t \leq 1$$

or

$$x(t) = t^2, y(t) = t^2, 0 \leq t \leq 1.$$

- ▶ Parametric equations of a line.

Velocity Vector (Tangent Vector)

- ▶ We define the velocity vector of the curve at t by:

$$\mathbf{x}'(t) = (x'_1(t), \dots, x'_n(t))$$

- ▶ If t represents time, then $x'_i(t)$ is the instantaneous velocity of the i -th coordinate along the curve at t .
- ▶ Consider $\mathbf{x}(t_0)$ as a vector in $\mathbb{R}^n$ with tail at $\mathbf{x}_0 = \mathbf{x}(t_0)$. Then $\mathbf{x}'(t_0)$ will be tangent to the curve at $\mathbf{x}_0$.

General Chain Rule (Theorem 14.4)

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $A : \mathbb{R}^s \rightarrow \mathbb{R}^n$ be C^1 functions. Let $\mathbf{x}^* = A(\mathbf{s}^*) \in \mathbb{R}^n$. Consider the composite function

$$H = F \circ A : \mathbb{R}^s \rightarrow \mathbb{R}^m.$$

Let $DF(\mathbf{x}^*)$ be the $m \times n$ Jacobian matrix of F at $\mathbf{x}^*$ and $DA(\mathbf{s}^*)$ be the $n \times s$ Jacobian matrix of A at $\mathbf{s}^*$. Then the Jacobian matrix of H is:

$$DH(\mathbf{s}^*) = DF(\mathbf{x}^*) \cdot DA(\mathbf{s}^*)$$

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- ▶ To compute the rate of change at a given point in any direction.
- ▶ $\mathbf{x} = \mathbf{x}^* + t\mathbf{v}$
- ▶ Evaluate F along the line: $g(t) = F(\mathbf{x}^* + t\mathbf{v}) = F(x_1^* + tv_1, \dots, x_n^* + tv_n)$

- Take derivative of g at 0:

$$\begin{aligned} g'(0) &= \frac{\partial F}{\partial x_1}(\mathbf{x}^*)v_1 + \cdots + \frac{\partial F}{\partial x_n}(\mathbf{x}^*)v_n \\ &= \left(\frac{\partial F}{\partial x_1}(\mathbf{x}^*), \dots, \frac{\partial F}{\partial x_n}(\mathbf{x}^*) \right) \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \\ &= DF_{\mathbf{x}^*} \cdot \mathbf{v} = \frac{\partial F}{\partial \mathbf{v}}(\mathbf{x}^*) = D_{\mathbf{v}}F(\mathbf{x}^*) \end{aligned}$$

- This is called the derivative of F at $\mathbf{x}^*$ in the direction of $\mathbf{v}$ or the directional derivative of F with respect to $\mathbf{v}$ at $\mathbf{x}^*$.

- ▶ The gradient vector

$$\nabla F_{\mathbf{x}^*} = \begin{pmatrix} \frac{\partial F}{\partial x_1}(\mathbf{x}^*) \\ \vdots \\ \frac{\partial F}{\partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

- ▶ Think of it as a vector in $\mathbb{R}^n$ with tail at $\mathbf{x}^*$.
- ▶ Consider only unit vector $\mathbf{v}$, $DF_{\mathbf{x}^*} \cdot \mathbf{v}$ measures rate of change from $\mathbf{x}^*$ in the direction $\mathbf{v}$.
- ▶ Example 14.9.

Theorem 14.2

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^1 function. At any point $\mathbf{x}$ at which $\nabla F(\mathbf{x}) \neq \mathbf{0}$, the gradient vector $\nabla F(\mathbf{x})$ points at $\mathbf{x}$ into the direction in which F increases most rapidly.

Example: Consider the production function

$$Q = F(K, L) = 4K^{3/4}L^{1/4}.$$

Current input bundles is (10,000, 625). In what proportions we should add K and L to increase the production most rapidly?

Solution: We compute the gradient vector of F at $(10,000, 625)$:

$$\nabla F(10,000, 625) = \begin{pmatrix} 1.5 \\ 8 \end{pmatrix}$$

So we deduce that we should add K and L at a ratio of 1.5 to 8.

- Exercise 14.18.

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- ▶ For $y = f(x_1, \dots, x_n)$, we define $\frac{\partial^2 f}{\partial x_j \partial x_i}$ with $i \neq j$ as the **cross/mixed partial derivatives**.
- ▶ C^1 : continuously differentiable, f' is continuous
- ▶ C^2 : twice continuously differentiable, f'' is continuous
- ▶ Suppose that $y = f(x_1, \dots, x_n)$ is C^2 in $\mathbb{R}^n$. then

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x})$$

We define Hessian matrix of f :

$$D^2 f_{\mathbf{x}^*} = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}^*) \end{pmatrix}$$

Example: Consider the general production function $Q = 4K^{3/4}L^{1/4}$. Then,

$$\frac{\partial Q}{\partial K} = 3K^{-1/4}L^{1/4}$$

$$\frac{\partial Q}{\partial L} = K^{3/4}L^{-3/4}$$

Then,

$$\frac{\partial^2 Q}{\partial L \partial K} = \frac{\partial}{\partial L} \left(\frac{\partial Q}{\partial K} \right) = \frac{\partial}{\partial L} \left(3K^{-1/4} L^{1/4} \right) = \frac{3}{4} K^{-1/4} L^{-3/4}$$

$$\frac{\partial^2 Q}{\partial K \partial L} = \frac{\partial}{\partial K} \left(\frac{\partial Q}{\partial L} \right) = \frac{\partial}{\partial K} \left(K^{3/4} L^{-3/4} \right) = \frac{3}{4} K^{-1/4} L^{-3/4}$$

$$\frac{\partial^2 Q}{\partial L^2} = \frac{\partial}{\partial L} \left(\frac{\partial Q}{\partial L} \right) = \frac{\partial}{\partial L} \left(K^{3/4} L^{-3/4} \right) = -\frac{3}{4} K^{3/4} L^{-7/4}$$

$$\frac{\partial^2 Q}{\partial K^2} = \frac{\partial}{\partial K} \left(\frac{\partial Q}{\partial K} \right) = \frac{\partial}{\partial K} \left(3K^{-1/4} L^{1/4} \right) = -\frac{3}{4} K^{-5/4} L^{1/4}$$

An economic application: law of diminishing marginal productivity.

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- Explicit functions: the endogenous variable is *explicitly* expressed as a function of x_i s

$$y = f(x_1, x_2, \dots, x_n)$$

- Implicit functions: for each $(x_1, x_2, \dots, x_n)$, the endogenous variable y is *implicitly* determined by

$$G(x_1, x_2, \dots, x_n, y) = 0$$

- 1 Given the implicit equation $G(x, y) = c$ and a point (x_0, y_0) such that $G(x_0, y_0) = c$, does there exist a continuous function $y = y(x)$ defined on an interval I about x_0 such that $G(x, y(x)) = c$ for all $x \in I$ and $y(x_0) = y_0$?
- 2 If $y(x)$ exists and differentiable, what is $y'(x_0)$?

Theorem 15.1

Let $G(x, y)$ be a C^1 function on a ball about (x_0, y_0) in $\mathbb{R}^2$. Suppose that $G(x_0, y_0) = c$ and consider the expression $G(x, y) = c$. If $\frac{\partial G}{\partial y}(x_0, y_0) \neq 0$, then there exists a C^1 function $y = y(x)$ defined on an interval I about the point x_0 such that :

(a) $G(x, y(x)) \equiv c$ for all x in I

(b) $y(x_0) = y_0$

(c) $y'(x_0) = -\frac{\frac{\partial G}{\partial x}(x_0, y_0)}{\frac{\partial G}{\partial y}(x_0, y_0)}$

Example

Consider the equation

$$G(x, y) \equiv x^2 - 3xy + y^3 - 7 = 0$$

At the point $(4, 3)$, one computes that

$$\frac{\partial G}{\partial x} = 2x - 3y = -1$$

$$\frac{\partial G}{\partial y} = -3x + 3y^2 = 15$$

$$y'(x_0) = -\frac{\frac{\partial G}{\partial x}(x_0, y_0)}{\frac{\partial G}{\partial y}(x_0, y_0)} = \frac{1}{15}$$

When $x_1 = 4.3$, then: $y_1 \approx y_0 + y'(x_0)\Delta x = 3 + (1/15) \times 0.3 = 3.02$

Theorem 15.2

Let $G(x_1, x_2, \dots, x_k, y)$ be a C^1 function around the point $(x_1^*, \dots, x_k^*, y^*)$. Suppose further that $(x_1^*, \dots, x_k^*, y^*)$ satisfies

$$\begin{aligned} G(x_1^*, \dots, x_k^*, y^*) &= c \\ \frac{\partial G}{\partial y}(x_1^*, \dots, x_k^*, y^*) &\neq 0 \end{aligned}$$

Then, there is a C^1 function $y = y(x_1, \dots, x_k)$ defined on an open ball B about $(x_1^*, \dots, x_k^*)$ so that

(a) $G(x_1, \dots, x_k, y(x_1, \dots, x_k)) \equiv c$ for all $(x_1, \dots, x_k)$ in B

(b) $y^* = y(x_1^*, \dots, x_k^*)$

(c) $\frac{\partial y}{\partial x_j}(x_1^*, \dots, x_k^*) = -\frac{\frac{\partial G}{\partial x_j}(x_1^*, \dots, x_k^*, y^*)}{\frac{\partial G}{\partial y}(x_1^*, \dots, x_k^*, y^*)}$

Implicit Function Theorem

Let F be a C^1 mapping of an open set $E \subset \mathbb{R}^{m+n}$ to $\mathbb{R}^m$ such that $F(y^*, x^*) = c^*$, where $y^* \in \mathbb{R}^m$ and $x^* \in \mathbb{R}^n$. Suppose $(\partial F / \partial y)(y^*, x^*)$ is invertible. Then,

- 1 there exist $\epsilon, \delta > 0$ such that for all $c \in B_\delta(c^*)$ and all $x \in B_\delta(x^*)$, there is a unique $y \in B_\epsilon(y^*)$ such that $F(y, x) = c$.
- 2 If we denote this y by $G(x, c)$, then G is C^1 and

$$\left(\frac{\partial G}{\partial x} \right) (x^*, c^*) = - \left(\frac{\partial F}{\partial y} \right)^{-1} (y^*, x^*) \left(\frac{\partial F}{\partial x} \right) (y^*, x^*).$$